In[ ]:=
```
Manipulate[(*Define the metric pressure gradient along xi*)
 metricPressure[xi_] := earlyInflation * Exp[-decayRate * xi] + currentAcceleration;
 (*Plot the geometric trace of a sub-luminal particle*)
 (*The xr position is dragged outward by the xi metric pressure*)
 Plot[metricPressure[xi] * (1 - beta^2), {xi, 0, 10}, PlotRange → {{0, 10}, {0, 50}},
  AxesLabel → {"Imaginary Inflation Axis (xi)", "Real Sub-luminal Distance (xr)"},
  PlotStyle → Thick], (*The Sliders*)
 {{earlyInflation, 10, "Early Universe Steepness (A)"}, 1, 20},
 {{decayRate, 1, "Inflation Drop-off Rate (k)"}, 0.1, 5},
 {{currentAcceleration, 2, "Present Day Acceleration (λ)"}, 0, 5},
 {{beta, 0.1, "Particle Velocity (β = v/c)"}, 0, 0.999}]
```

Out[ ]=

